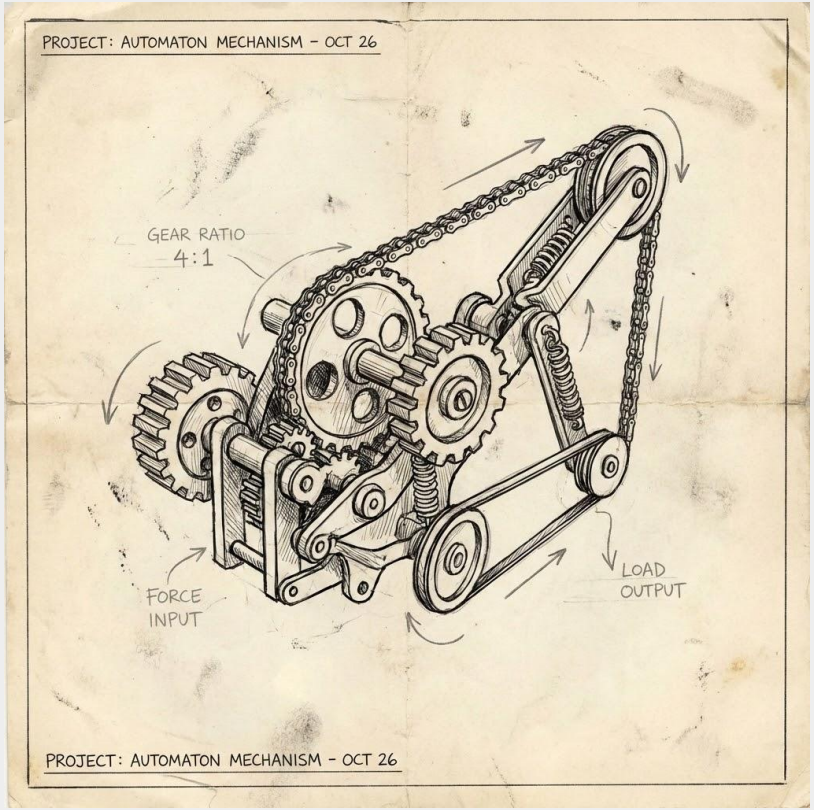
CSC3041F

Control

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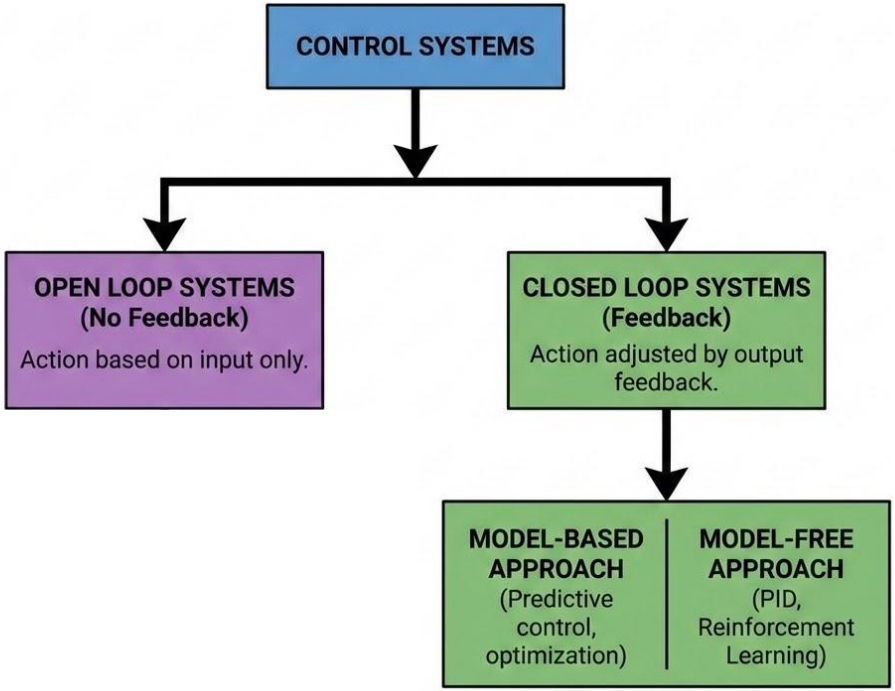


Image: Lucie de Moyencourt



Control Systems

CONTROL SYSTEMS RELATIONSHIPS

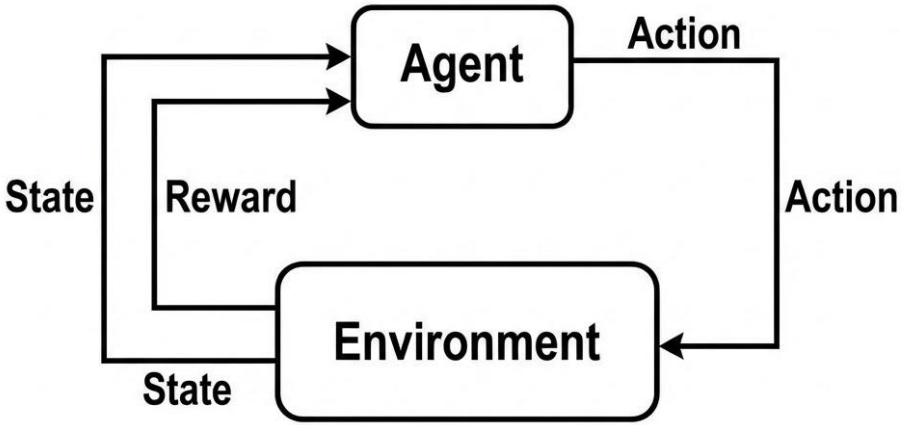


Plant Characteristics

- Complexity?
  - Linear or non-linear
  - Does superposition apply?
- Do we have a model of the plant?
  - Human intuition or physics can be used? Or are we interacting with an environment?
- Can the external forces be encapsulated in the plant model?
- Is the task repetitive or is their uncertainty?
- Number of inputs/outputs?

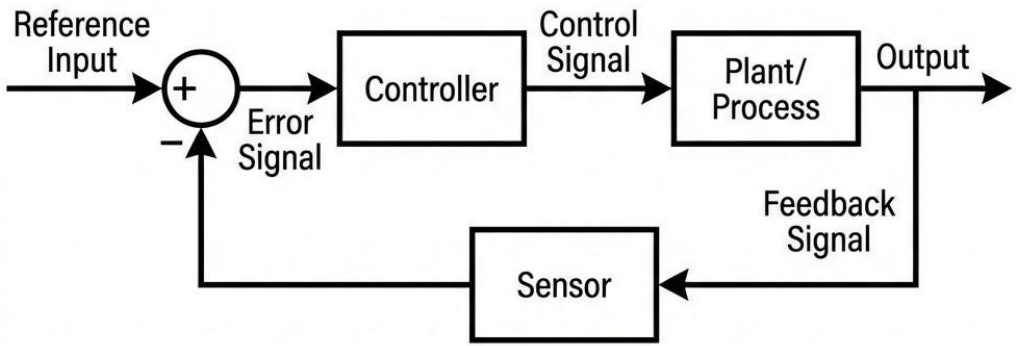
Reinforcement Learning

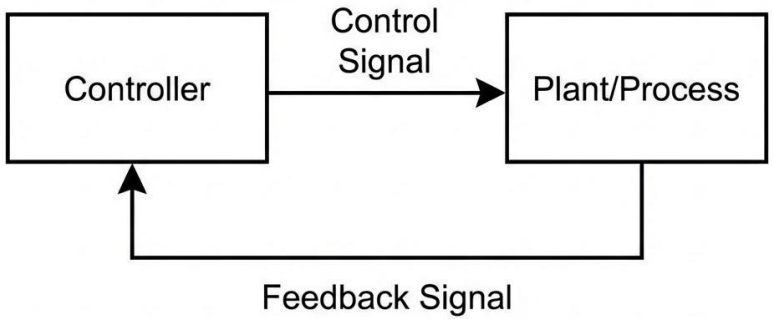




Reinforcement Learning Feedback Loop

Feedback Control System

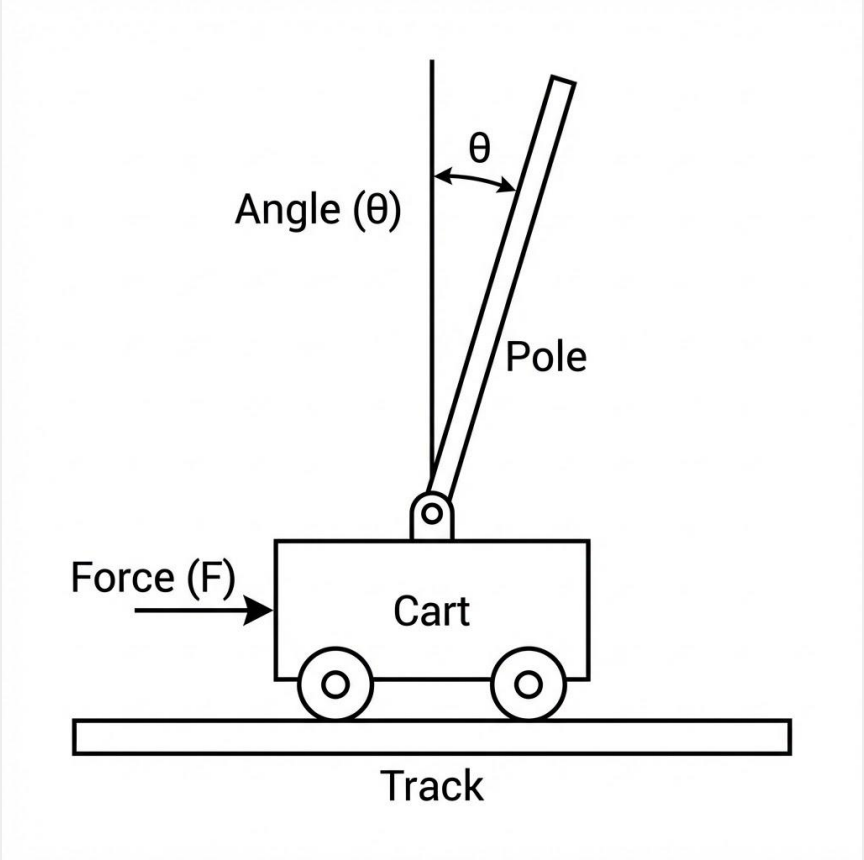




PID to Reinforcement Learning

	PID	**Reinforcement Learning**
SISO vs MIMO	SISO	MIMO
Linear vs Non-Linear	Linear	Non-Linear
Adaptability	Fixed	Can continue to learn ‘online’

Cartpole



Feature	PID Controller	RL Agent (PPO)
Foundations	Calculus & Linear Algebra.	Probability & Optimisation.
Initial Performance	Perfectly smooth (if tuned well).	Often "jittery" or sub-optimal.
Reaction to 2× Mass	Brittle: Likely to fail or oscillate.	Robust: Could generalise to new dynamics.
Debugging	Easy (Check the roots of the equation).	Difficult (Black-box neural network).

Automation vs Autonomy

Automation vs Autonomy

- Automation (Classical Control): A system that follows a pre-defined set of rules to perform a repetitive task (e.g., a robotic arm on an assembly line).

- Autonomy (RL-Based Control): A system that can adapt its behaviour to achieve a goal in an uncertain or changing environment (e.g., a drone navigating a construction site).

Integration Strategies: How RL Meets Hardware

1. RL as the Direct Controller (End-to-End)
    a. The RL agent replaces the PID loop entirely. It observes raw sensor data (pixels, joint angles) and outputs raw control signals (voltages, torques).

    b. +: Can learn highly complex, non-linear movements that humans can't easily model.

    c. -: Hard to guarantee safety; "Black Box" nature makes it risky for heavy machinery.

Integration Strategies: How RL Meets Hardware

2. Hierarchical Control (The Supervisor Model)

a. Low-Level: Traditional PID controllers handle the high-speed "reflexes" (keeping motors at the right RPM).

b. High-Level: An RL Agent acts as the "Manager," sending target set-points to the PID loops every few milliseconds.

c. Example: In a self-driving car, RL decides the "Target Steering Angle," while a PID controller ensures the wheels actually turn to that exact angle.

Integration Strategies: How RL Meets Hardware

3. RL for System Identification

    a. The RL agent is used to "learn the model" of a complex system (A and B matrices) that is too difficult to derive by hand. Once the model is learned, we can use classical Model Predictive Control (MPC).

The Implementation Pipeline

1. Simulation

2. Domain randomisation

3. Deployment

Real-World Applications in South Africa

1. Mining: Autonomous underground vehicles (LHDs) navigating tunnels using RL-based LIDAR processing.

2. Energy (Eskom context): Using RL to optimise the load balancing of a micro-grid when integrating renewable sources like wind and solar, which are notoriously non-linear and unpredictable.

3. Agriculture: RL-controlled drones for precision spraying in orchards, adjusting for wind speed (disturbance rejection) in real-time.

The "Safety" Elephant in the Room

- In classical control, we can prove Stability mathematically. In Deep RL, we usually can't.
  - The Challenge: How do we stop an RL agent from "exploring" a state that breaks the machine?
  - The Solution: Safe RL. We use "Shielding" or "Safety Layers" that sit between the RL agent and the motor. If the agent suggests an action that is physically dangerous, the safety layer overrides it.